

Total No. of Questions : 4]

SEAT No. :

PE26

[6579]-327

[Total No. of Pages : 1

T.E. (Computer Engg./Computer Science) (Insem)
SYSTEMS PROGRAMMING AND OPERATING SYSTEM
(2019 Pattern) (Semester - I) (310243)

Time : 1 Hour]

[Max. Marks : 30

Instructions to the candidates:

- 1) Answer Q.1 or Q.2, Q.3 or Q.4.
- 2) Neat diagrams must be drawn wherever necessary.
- 3) Figures to the right indicate full marks.
- 4) Assume suitable data, if necessary.

- Q1)** a) Explain Different assembly language statements with examples? [7]
b) Explain Pass I and Pass II of two pass Assemblers with suitable examples and flowcharts? [8]

OR

- Q2)** a) Explain different data structures associated with PASS 1 & PASS 2 Assembler with example and structure? [8]
b) What is System Programming? Explain Foundation of System Programming? [7]

- Q3)** a) Explain in detail the various phases of a compiler, describes the output for the following Expression after each phase. $A = b * c + c * d$ [7]
b) What are macro processors? Explain Two Pass Macro with a suitable example and algorithm? [8]

OR

- Q4)** a) Explain different argument passing mechanism in Macros with an example? [7]
b) Write Short note on: [8]
i) Compiler and interpreter
ii) Macro and Assembler

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SEAT No. :

PC-27

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[6360]-27

T.E. (Computer Engineering) (Insem)
SYSTEMS PROGRAMMING AND OPERATING SYSTEM
(2019 Pattern) (Semester - I) (310243)

Time : 1 Hour]

[Max. Marks : 30

Instructions to the candidates :

- 1) *Answer Q1 or Q2, Q3 or Q4.*
- 2) *Neat Diagrams must be drawn wherever necessary.*
- 3) *Figure to the right indicate full marks.*
- 4) *Assume suitable data if necessary.*

Q1) a) Draw and explain flowchart of pass I of two pass assembler with example. [8]

- b) Differentiate : [7]
- i) Literal and Immediate operand.
 - ii) Assembler and Compiler

OR

Q2) a) Consider following Assembly code and show output of pass-I of two pass Assembler with entries in Mnemonic Opcode Table, Pseudo Opcode Table, Symbol Table, Literal Table and Pool Table. [8]

```
PROG START 50
USING PROG+2, 15
L1, FIVE
AL, = F '2'
LTORG
ST 1, RES
FIVE DC F '4'
RES DS F '4'
RES DS IF
END
```

- b) Enlist and explain necessity of different data structures used in Pass-I of two pass Assembler? [7]

P.T.O.

- Q3)** a) Explain the Phases of Compiler and their output with an example. [8]
b) Define macro. What are the advantages of macro facility? How they are different from function. [7]

OR

- Q4)** a) What is the use of AIF and AGO pseudo-op in macro? Explain macro expansion with positional parameter with the help of suitable example. [8]
b) Explain the concept of single pass Macro processor with example. Give example for macro calls within the macro. [7]

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SEAT No. :

P8557

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Oct-22/TE/Insem - 527

T.E. (Computer Engineering)

SYSTEMS PROGRAMMING AND OPERATING SYSTEM

(2019 Pattern) (Semester - I) (310243)

Time : 1 Hour]

[Max. Marks : 30

Instructions to the candidates:

- 1) *Neat diagrams must be drawn wherever necessary.*
- 2) *Figures to the right indicate full marks.*
- 3) *Assume suitable data, if necessary.*

- Q1)** a) Draw and explain flowchart of Pass-I of two pass assembler with suitable example. [8]
- b) Differentiate between literal and immediate operand. [7]

OR

- Q2)** a) Explain in detail with a suitable example, the formats and contents of databases used in assembler design. [8]
- b) Explain algorithm of pass 1 of two pass assembler. [7]
- Q3)** a) Define Macro. What are the advantages of macro facility? How they are different from functions? [8]
- b) Write Short note on [7]
- i) Compiler
 - ii) Interpreter

OR

- Q4)** a) What are different data structures required for Two Pass Macro Processor? Justify which data structures are implemented at that time of processing Macro definition, Macro call and Macro Expansion. [8]
- b) Explain argument passing mechanism in Macros with suitable examples? [7]

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SEAT No. :

P-5027

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[6187]-427

T.E. (Computer Engineering/(A.I.D.S)) (Insem.)
SYSTEMS PROGRAMMING AND OPERATING SYSTEM
(2019 Pattern) (Semester - I) (310243)

Time : 1 Hour]

[Max. Marks : 30

Instructions to the candidates :

- 1) Attempt Q.No. 1 or Q.No. 2, and Q.No. 3 or Q.No. 4.
- 2) Neat diagrams must be drawn wherever necessary.
- 3) Figures to the right indicate full marks.
- 4) Assume suitable data, if necessary.

Q1) a) What is purpose of Assembler pass 1? Draw and explain overview of Assembler pass 1 flow chart [8]

b) Compare system softwares with Application softwares? Explain benefits of Assembly Language. [7]

OR

Q2) a) Discuss need of intermediate code of assembly program. Generate intermediate code for an assembly language program given in Question 2b using any one variant of intermediate code. [8]

b) Explain the output of pass-I of two pass Assembler with respect to the given program: [7]

```
START      600
READ A
READ B
LOOP  MOVER    AREG, A
      MOVER    CREG, B
      SUB AREG,='I'
      BC  GT,LOOP
      STOP
A      DS      1
B      DS      2
END
```

P.T.O.

Q3) a) What is Macro? Explain Macro definition, Macro Call and Macro Expansion with an example. **[8]**

b) Differentiate **[7]**

i) Macro and subroutine

ii) Compiler and Interpreter

OR

Q4) a) Explain various phases of the Compiler for the expression $x = I + R * 60$ where the data type of R is float. **[8]**

b) Explain briefly the algorithm of pass I of two pass macro processor? **[7]**

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